

GUIDELINES FOR “DATA & METHODS” INTERIM PRESENTATION AND REPORT

A. PRESENTATION FORMAT. Your group should prepare a 15 minute talk to be presented in class on Thursday March 27th. Slides should be prepared and presented in PowerPoint (or equivalent), i.e., not using the whiteboard or Doc Cam in our classroom. While not all members of your group need to deliver your talk, it’s advisable to have more than one group member participate in your group’s presentation.

B. PRESENTATION CONTENT. In your group’s presentation, be sure to include a few slides each, addressing the following:

1. The motivating application and some primary questions of interest.
2. A summary of the types of data sets you’re analyzing using data science methodology.
3. A brief discussion of techniques used to clean and normalize the data.
4. A summary and brief technical description of the methods and algorithms that you’re currently using.
5. A summary of steps completed, current work, next steps and, if available, one example of an initial result.
6. One slide with a list of the software tools, packages, and algorithms that you’ve used so far in your project. This should include tools used to prepare your presentation and report.

C. WRITTEN REPORT. Your group should prepare a written interim report focusing on the data and methods involved in your project to date. In contrast to the items listed in Sec. B, the emphasis is to start building and refining content for your final project report. To this end, your report should contain the following three sections.

1. **Background and Introduction.** Qualitatively describe the motivation for the problem you’re tackling this semester, some questions you’re aiming to address, a few initial or anticipated methodological approaches, and the types of data sets that are driving your efforts. This section should be written as a series of paragraphs with well-flowing sentences, i.e. it’s best not to break this part of your report into subsections or to include any equations.
2. **Data Sets.** Describe, with some technical details, the data sets available to you, their sources, how you’re cleaning and normalizing them, and which data subsets are being analyzed using your methodological approaches. Please see the lecture notes on *Fundamentals of Learning (Topic I)* from FDS 610 for sample notation to use.
3. **Methods and Algorithms.** Describe and outline, with some technical details, the methods and/or algorithms being used in your project (to date). As much as possible, avoid “black-box” descriptions, providing some details regarding the methodology being employed at each stage of your workflow and why you chose it. When describing the use of specific methods or algorithms be sure to indicate what is required as input and

what is “received” as output.

Due Date. One report per group must be submitted via Moodle by **11:59pm on Monday March 31st**. The preferred format is a single pdf file. You should also submit a copy of the slides from your presentation, either as a single pdf or PowerPoint file.

Some general tips and advice:

- You should prepare your report using either LaTeX or Microsoft Word. For the former, note that NC State has an Overleaf site license. If using Microsoft Word, please use the built-in equation editor or LaTeXIt to typeset your equations. Please contact the instructor if you wish to use alternate software to prepare your report.
- Inclusion of code or code-based “notebooks” (jupyter) in this report is entirely OPTIONAL. If you wish to include such items in this report, e.g., to save you time in future reports, please place them in an Appendix. You can cite the Appendix as needed in sections 2 and 3 of your report.
- Your report should include a few illustrative figures, with captions.
- Please format your report single-spaced, using a font size between 10pt and 12pt. A report that is 5 pages in length (not including appendices) is too short. A report that is 20 pages in length (not including appendices) is too long.
- DO NOT use generative AI tools to aid in preparing this report (an explanation will be provided in an upcoming class).
- Some **samples of final reports** analogous to those expected at the end the course will be provided on our course Moodle page and reviewed in an upcoming class.